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BACHELOR THESIS

Time-dependent characterization and constitutive modeling of 3D-printed polymers

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Abstract

Tango Black is a polymer with many applications in medical devices, consumer electronics and testing. The constitutive modeling of such an important material allows to predict the behavior of structures and to run simulations of experimental testing. In this contribution, many tensile, loading-unloading and single-step relaxation tests of Tango Black samples have been realized. To model the data obtained, we use a model for isotropic incompressible elastomers proposed by Lopez-Pamies [7]. To validate the fitted model, we observe the capacity of the model to reproduce experimental results.

In the following, we present the experimental results of the many mechanical tests realized and the constitutive modeling of Tango Black based on the previous experimental data. The experimental procedure showcases the different parameters of the tests as well as the challenges faced and how we overcame them by finding diverse solutions. Finally, an application of the constitutive modeling of Tango Black will be presented: the Compression of Tango Black Hexagonal Structures.

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1 Introduction

Tango Black is a material highly used in testing applications. Thus, it is useful to model this material which is necessary to run simulations to understand how diverse structures behave under forces and to compare them with experimental data. Therefore, we will be doing the experimental characterization under uni-axial tension and cyclic loads of a 3D-printed polymer, Tango Black. Relaxation tests will be carried out too.

In addition, to a more classical dog-bone type specimen, a new specimen geometry, that of a band, will be considered too. The difficulty with this type of material is its semi brittle nature leading to fracture at stretches in the order of 40-50% and sensitivity at the hold position. The main aim is to study the response of the polymer under cyclic loads below the fracture strain and at different rates to reveal the underlying non-linear viscoelastic behaviour. The study will be supplemented with the constitutive modeling of the polymer using available theoretical models for such cases like the Neo-Hookean incompressible model, isotropic incompressible elastomers model proposed by Lopez-Pamies [7] ...

In the following, we start by presenting the main theoretical notions related to viscoelasticity and tensile tests. Then, we show how we designed our sample and setup, using SolidWorks simulations, and the diverse experimental techniques used: Tensile Test, DIC (Digital Image Correlation) and 3D printing. We expose the results and fit the data with a convenient model for elastomers. Finally, we display one application of modeling Tango Black: Compression of Tango Black Hexagonal Structures.



FIGURE 1.1: Tango Black.

2 Theoretical Background

In order to better understand the experimental and numerical work undertaken, a short theoretical background about polymers, tensile test and viscoelasticity will be presented in this section.

2.1 Tango Black, a polymer

A polymer is a material composed from very long molecular chains containing many repeating sub-units, monomers. Due to this molecular composition, it is characterized by high elasticity, viscoelasticity and toughness.

Tango Black is a polymer from the family of Tango. It is characterized by its brittle nature. It has various industry applications like Automotive Components, Computer Components, Medical Devices and Consumer Electronics. It is particularly used for shoes, handles, seals, knobs and rapid tooling. [3]

Tango Black is considered as an incompressible material since its Poisson ratio is 0.49, which will be measured in [Section 3.3](#).

TABLE 2.1: Mechanical characteristics of Tango Black given by Stratasys [2].

	ASTM	Metric
Tensile Strength	D-412	1.8-2.4 MPa
Elongation at Break	D-412	45-55%
Compressive Set	D-395	0.5-1.5%
Shore Hardness (A)	D-2240	60-62 Scale A
Tensile Tear Resistance	D-624	3-5 Kg/cm
Polymerized density	D-792	1.14-1.15 g/cm ³

2.2 Viscoelasticity

2.2.1 Definition

Viscoelastic material behavior is a combination of elastic and viscous behavior. Part of the deformation energy will be dissipated, while the rest is stored as reversible elastic energy. Viscoelastic behavior can be characterized by mechanical models build from springs and dash-pots. We are interested in non-linear viscoelasticity i.e. viscoelasticity at high strain.

To investigate the viscoelastic behavior of a material tensile tests, cyclic tensile tests and relaxation tests are performed at different strain rates. [7]

2.2.2 Viscoelasticity of Polymers

Polymers are considered as a viscoelastic material. In fact, we can find entanglement points between polymer chains at its initial equilibrium state. Once stretched at small strain, we have two phenomena which will occur: chain straightening and bond stretching, they will result in a non equilibrium state of the polymer which once unloaded will go back to its original shape. This is the elastic behavior of polymers. But, once stretched at higher strains, the entanglement points break, which will allow the chains to slip.

This is one of the dissipating mechanisms of polymers. When unloaded the polymer will have achieved a new equilibrium state with its polymer chains having reached a new configuration with less entanglement points. This is the viscous behavior of polymers.

2.2.3 Neo-Hookean incompressible model

The equilibrium and non-equilibrium free energy functions associated with the Neo-Hookean incompressible model for hyperelastic material are:

$$\psi^{\text{Eq}}(\mathbf{F}) = \frac{\mu_1}{2} (\mathbf{F} \cdot \mathbf{F} - 3) \text{ and } \psi^{\text{NEq}}(\mathbf{F}^e) = \frac{\mu_2}{2} (\mathbf{F}^e \cdot \mathbf{F}^e - 3) \quad (2.1)$$

with $\mathbf{F}$ the the deformation gradient tensor.

We are interested in the uni-axial displacement control loading. The boundary conditions would be:

$$\mathbf{S} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & S_{33} \end{pmatrix} \quad \mathbf{F} = \begin{pmatrix} \frac{1}{\sqrt{\lambda}} & 0 & 0 \\ 0 & \frac{1}{\sqrt{\lambda}} & 0 \\ 0 & 0 & \lambda \end{pmatrix} \quad \mathbf{F}^v = \begin{pmatrix} \frac{1}{\sqrt{\lambda_v}} & 0 & 0 \\ 0 & \frac{1}{\sqrt{\lambda_v}} & 0 \\ 0 & 0 & \lambda_v \end{pmatrix} \quad (2.2)$$

After derivation of the equations for the uni-axial tension, we get the following stress function S along the vertical direction:

$$S = \mu_1 \left(\lambda - \frac{1}{\lambda^2} \right) + \mu_2 \left(\frac{\lambda}{\lambda_v^2} - \frac{\lambda_v}{\lambda^2} \right) \quad (2.3)$$

And the following ODE in λ_v :

$$\dot{\lambda}_v = \frac{\mu_2}{3\eta_k} \left(\frac{\lambda^3 - \lambda_v^3}{\lambda \lambda_v} \right) \quad (2.4)$$

2.2.4 Model for isotropic incompressible elastomers

This model for isotropic incompressible elastomers uses the two-potential constitutive framework — also known as the “generalized standard materials” framework to rubber viscoelasticity. This model is consistent with the material frame indifference, material symmetry, and the second law of thermodynamics. [7]

The goal of the two-potential constitutive framework is to describe how a material stores and dissipates energy by using two thermodynamic potentials: (i) a free energy function ψ and (ii) a dissipation potential ϕ .

$$\psi = \psi^{\text{Eq}}(\mathbf{F}) + \psi^{\text{NEq}}(\mathbf{F}, \mathbf{F}^v) \quad \text{and} \quad \phi = \phi(\mathbf{F}, \mathbf{F}^v, \dot{\mathbf{F}}^v) \quad (2.5)$$

where the free energy function ψ^{Eq} serves to characterize the thermodynamic equilibrium states of the elastomer, while ψ^{NEq} serves to account for the additional energy storage at non-equilibrium states.

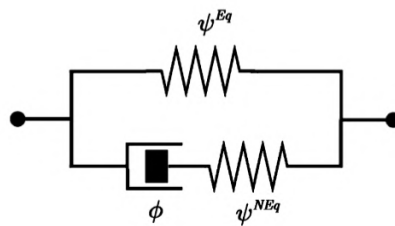


FIGURE 2.1: Rheological model of rubber viscoelastic behavior [7].

As we see in Figure 2.2, we assume the existence of two maps that divide the deformation of the body into an elastic part χ_e and a viscous part χ_v .

$$\chi(X) = \chi_e \circ \chi_v = \chi_e(\chi_v(X)) = \chi_e(\xi) \quad \text{with} \quad \xi = \chi_v(X) \quad (2.6)$$

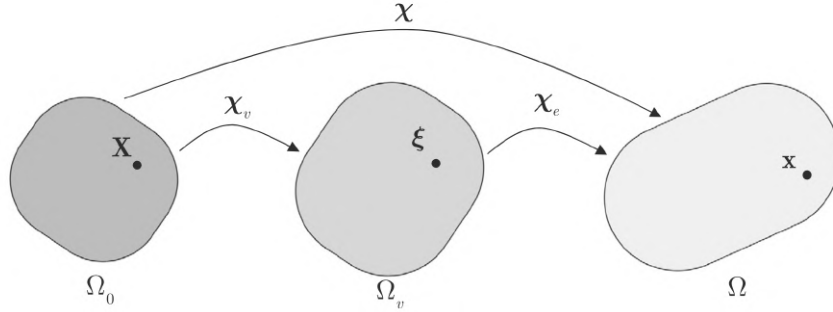


FIGURE 2.2: Schematic of the deformation field χ , mapping material points $\mathbf{X}$ in the undeformed reference configuration Ω_0 to their spatial position $\mathbf{x}$ in the current deformed configuration Ω , and its assumed composition [7].

Then, we get that the deformation gradient tensor $\mathbf{F}$, can be decomposed into an elastic part $\mathbf{F}^e$ and a viscous part $\mathbf{F}^v$:

$$\mathbf{F} = \frac{\partial \chi}{\partial \mathbf{X}} = \frac{\partial \chi_e}{\partial \xi} \frac{\partial \chi_v}{\partial \mathbf{X}} = \mathbf{F}^e \mathbf{F}^v \quad (2.7)$$

Now, we choose the following equilibrium and non-equilibrium free energy functions, proposed by Lopez-Pamies [7]

$$\psi^{\text{Eq}}(\mathbf{F}) = \begin{cases} \frac{3^{1-\alpha_1}}{2\alpha_1} \mu_1 [I_1^{\alpha_1} - 3^{\alpha_1}] + \frac{3^{1-\alpha_2}}{2\alpha_2} \mu_2 [I_1^{\alpha_2} - 3^{\alpha_2}] & \text{if } J = 1 \\ +\infty & \text{otherwise} \end{cases} \quad (2.8)$$

$$\psi^{\text{NEq}}(\mathbf{F}^e) = \begin{cases} \frac{3^{1-a_1}}{2a_1} m_1 [I_1^{a_1} - 3^{a_1}] + \frac{3^{1-a_2}}{2a_2} m_2 [I_1^{a_2} - 3^{a_2}] & \text{if } J = 1 \\ +\infty & \text{otherwise} \end{cases} \quad (2.9)$$

We are interested in the uni-axial displacement control loading. The boundary conditions would be:

$$\mathbf{S} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & S_{33} \end{pmatrix} \quad \mathbf{F} = \begin{pmatrix} \frac{1}{\sqrt{\lambda}} & 0 & 0 \\ 0 & \frac{1}{\sqrt{\lambda}} & 0 \\ 0 & 0 & \lambda \end{pmatrix} \quad \mathbf{F}^v = \begin{pmatrix} \frac{1}{\sqrt{\lambda_v}} & 0 & 0 \\ 0 & \frac{1}{\sqrt{\lambda_v}} & 0 \\ 0 & 0 & \lambda_v \end{pmatrix} \quad (2.10)$$

After derivation of the equations for the uni-axial tension (Appendix A), we get the following stress function S along the vertical direction:

$$S = \left[\sum_{r=1}^2 3^{1-\alpha_r} \mu_r \left(\frac{2}{\lambda} + \lambda^2 \right)^{\alpha_r-1} \right] \left(\lambda - \frac{1}{\lambda^2} \right) + \left[\sum_{r=1}^2 3^{1-a_r} m_r \left(\frac{2\lambda_v}{\lambda} + \frac{\lambda^2}{\lambda_v^2} \right)^{a_r-1} \right] \left(\frac{\lambda}{\lambda_v^2} - \frac{\lambda_v}{\lambda^2} \right) \quad (2.11)$$

And the following ODE in λ_v :

$$\dot{\lambda}_v = \frac{\sum_{r=1}^2 3^{1-a_r} m_r \left(\frac{2\lambda_v}{\lambda} + \frac{\lambda^2}{\lambda_v^2} \right)^{a_r-1}}{3\eta_k} \left(\frac{\lambda^2}{\lambda_v} - \frac{\lambda_v^2}{\lambda} \right) \quad (2.12)$$

with

$$\eta_k = \eta_\infty + \frac{\eta_0 - \eta_\infty + k_1 \left[\left(\frac{2}{\lambda_v} + \lambda_v^2 \right)^{\beta_1} - 3^{\beta_1} \right]}{1 + \left(k_2 J_2^{NEq} \right)^{\beta_2}} \quad (2.13)$$

and

$$J_2^{NEq} = \frac{1}{3} \left(\frac{\lambda_v}{\lambda} - \frac{\lambda^2}{\lambda_v^2} \right)^2 \left(\sum_{r=1}^2 3^{1-a_r} m_r \left(\frac{2\lambda_v}{\lambda} + \frac{\lambda^2}{\lambda_v^2} \right)^{a_r-1} \right)^2 \quad (2.14)$$

2.3 Tensile Test

Tensile testing is the application of axial forces on a specimen with known dimensions until fracture. The result of such test is a stress-strain curve. The axial forces will generate internal forces and thus, normal stresses in the specimen which will lead to normal strain and thus, elastic and then plastic deformation. This test allows to determine many material properties like yield strength, ultimate tensile strength, percentage elongation at fracture and many others which can help us to describe the strength and ductility of the materials.

We define some important quantities:

The engineering stress:

$$S = \frac{F}{A_0}$$

With F the applied force, A_0 the initial cross section area of the specimen.

The engineering strain:

$$\lambda - 1 = \frac{u}{L}$$

With u the change in length due to the applied force, L the initial length of the specimen.

3 Characterisation and constitutive modeling of Tango Black

In this section, we will first present the experimental setup and experimental methods used in our mechanical testings. Then, we will discuss the obtained results and fit them with a model for hyperelastic elastomers. We will end this section with a fracture study of Tango Black.

3.1 Design of the sample

3D printing or additive manufacturing is the process of building a three dimensional structure from a CAD model. This computer controlled process consists in taking a material (plastics, liquids, metals...), heating it to make it liquid and then placing it, layer by layer, on a tray where it will solidify. The movement of the head of the printer is controlled by a CAD (Computer aided design) file which is generally written in G-code. It's role is to guide the horizontal and vertical movement of the printer by giving it coordinates along the X, Y and Z axes.

To produce the samples we tested, we used the 3D printer: "Objet Eden 260VS". It has a precision of 16 microns and can use 14 different materials, like Vero White Plus, Tango Black or Tango Black Plus... [2] It used UV-light to solidify the material, which was in our case Tango Black.



FIGURE 3.1: 3D Printer Objet Eden 260VS [1].

We used Vero White Plus (RDG835) to build parts of the experimental setup, Tango Black (FLX973) for our samples and Resin Support (SUP705) for the support used by the printer.

Through this project, we had to change the geometry of our sample many times depending on the problems we were facing (Figure 5 Appendix D).

The first geometry used was a dog bone. Clamps were used to realize tensile tests. The problem faced was that when the tensile tests were realized, the sample broke in the non-linear part of the dog bone or slipped out of the clamps.

From here, we thought of changing the setup to grips which will hold the sample to perform tensile tests. We chose the geometry ① (Figure 3.2) and simulated a tensile test of this geometry on SolidWorks. We can see the results of this simulation in Figure 3.3. We observe that we have the stress concentration localized in the uniform length of the sample. Furthermore, we see that the maximal stress is reached at the tangent point between the uniform length and the curvature (red arrow), which will end up being the point of fracture of the sample.

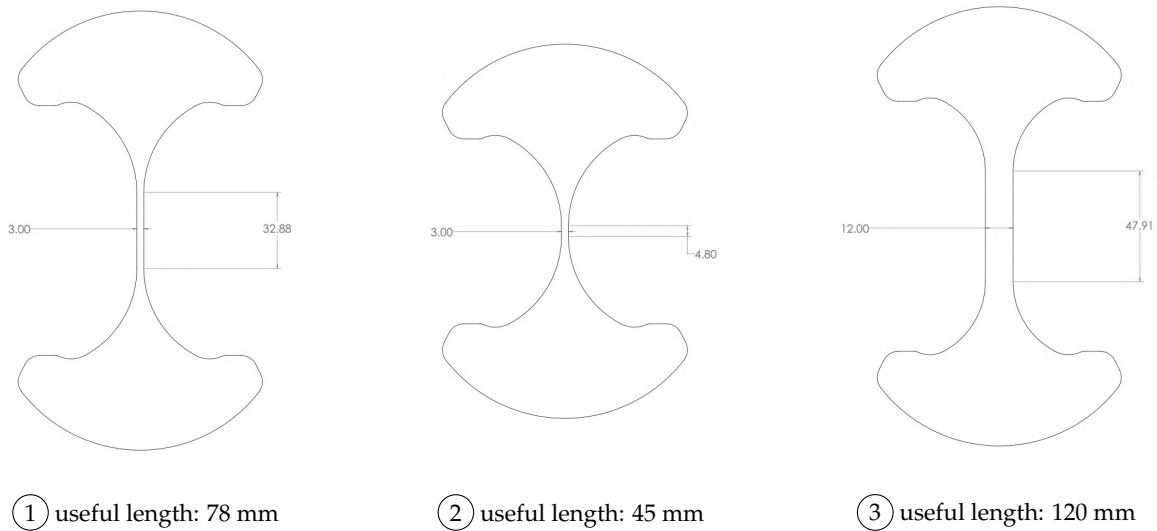


FIGURE 3.2: Samples used in the mechanical testing, with different cross sectional areas and uniform lengths, thickness: 3 mm.

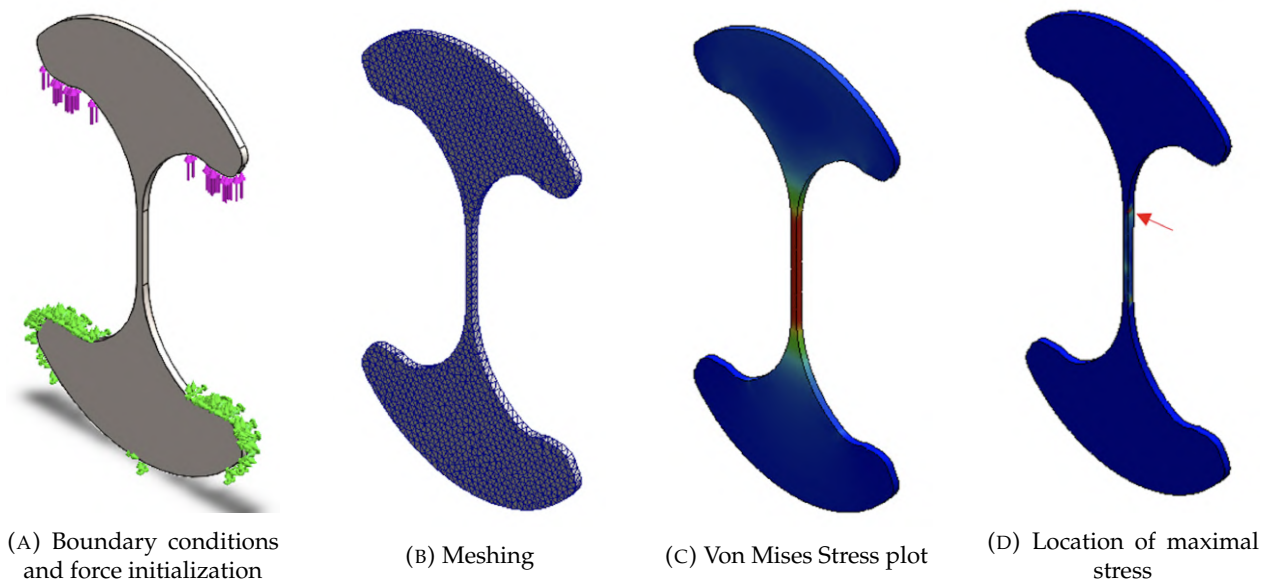
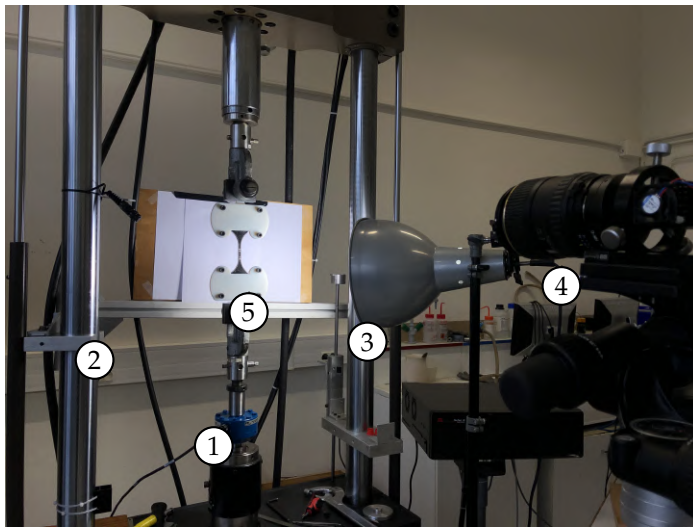


FIGURE 3.3: Simulation of a tensile test of the geometry used in the tests realized on SolidWorks.

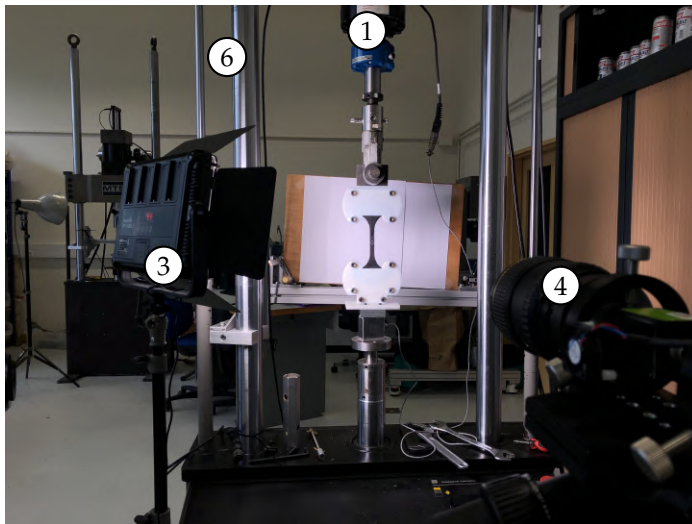
3.2 Experiments



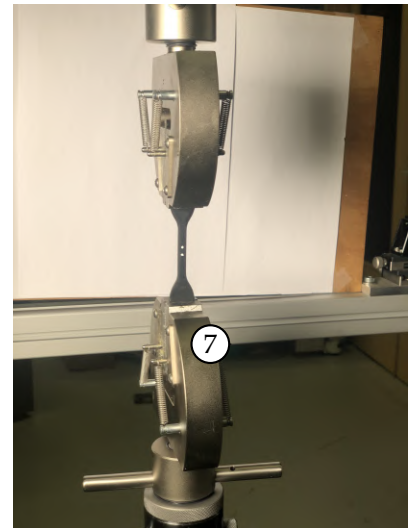
(A) Tensile Test setup using grips



(B) Tensile Test setup using grips (Close view)



(C) Tensile Test setup using grips for relaxation



(D) Tensile Test setup using clamps

FIGURE 3.4: Experimental setup for tensile, cyclic and relaxation testing, with DIC strain measurement. 1. Load Cell, 2. MTS 250kN, 3. Lamp, 4. Camera, 5. Grips, 6. MTS 100kN, 7. Clamps.

3.2.1 Experimental Setup

The tensile experiments were realized under displacement control in a hydraulic testing machine. Cyclic tests were conducted under constant strain rate, while relaxation tests were conducted under constant deformation level. After printing the samples, we first clean them from the support of the printer. Then, we place them in the grips, on which we add two plates bolted to the grips to avoid the buckling of the sample.

The machine was controlled by softwares where we input the parameters of the test. For tensile tests, we used the Station Manager Software (Figure 3 Appendix B) where we initialize the displacement rate and the ending criterion. For Cyclic and relaxation tests, we used TestSuite Multipurpose Elite, which is a software where we can give a list of commands to be realized by the machine. The program used for

loading-unloading tests can be seen in Figure 2 (A) Appendix B and the one for relaxation tests can be seen in Figure 2 (B) Appendix B.

For the data acquisition, we recorded the required load, using a load cell, every unit of time depending on the expected length of the experiment. The strain measurement was realized by DIC (Digital Image Correlation) (Figure 3.5). The data was collected using the Software "Sylvie IC" (Figure 1 Appendix B). A smoothing technique was used to clean up the data on Python. Tensile and Cyclic Tests were fit by polynomials, while Relaxation tests were fit by decaying exponential. All stress calculations are expressed as nominal or engineering stress and the strain calculations are expressed as nominal or engineering strain. The strain rate is defined as $\dot{\Delta L}/L_0$ where L_0 is the useful length, measured experimentally, and ΔL the length difference between the final and initial length of the specimen.

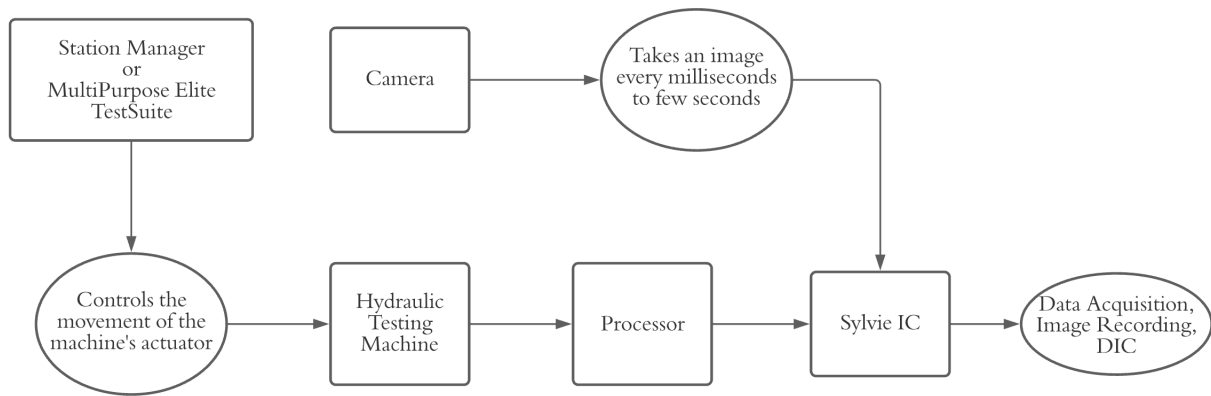


FIGURE 3.5: Block diagram of experimental setup.

3.2.2 DIC for Strain measurement

DIC or Digital Image Correlation is a strain measurement technique. It consists of a camera directed towards the sample which takes images at a specific interval of time ranging from milliseconds to hundred of seconds. DIC takes this sequence of pictures and by tracking blocks of pixels it is able to calculate displacement and thus, gives strain plots. It can then compute the average strain to give the local strain. It requires to add fingerprints points on the surface of the sample to make this process possible. These points need to be placed in the middle of the uniform length.

3.2.3 Hydraulic Tensile Testing Machines

In order to complete all the experiments, we used hydraulic tensile testing machines MTS 100kN and MTS 250kN.

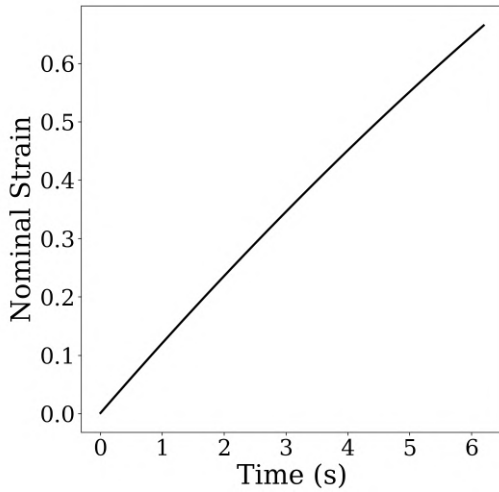
TABLE 3.1: Characteristics of the MTS Tensile Testing machines.

Characteristics	MTS 100kN	MTS 250kN
Load range	± 100 kN	± 250 kN
Stroke	± 100 mm	± 80 mm
Actuator maximum speed	100 mm/s	30 mm/s

3.2.4 Tensile and Cyclic Testing

To realize Tensile and Cyclic tests of Tango Black we used the geometry ① (Figure 3.2). Nonetheless, since the tensile machine MTS 250kN had a limit of displacement rate of 30 mm/s, we had to use MTS

100kN, where the maximal displacement rate possible was 100 mm/s. This allowed us to increase our strain rate from: $\frac{30\text{mm/s}}{78\text{mm}} \approx 0.4\text{s}^{-1}$ to $\frac{100\text{mm/s}}{78\text{mm}} \approx 1.3\text{s}^{-1}$. Furthermore, we wanted to achieve a higher strain rate, to observe a higher dissipation due to viscoelasticity. Therefore, we decreased the useful length of our geometry. So we got a new geometry: geometry ② (Figure 3.2). With this geometry we managed to reach a strain rate of $\frac{100\text{mm/s}}{45\text{mm}} \approx 2.2\text{s}^{-1}$ and thus get a bigger dissipation due to viscoelasticity. Moreover, one limitation faced while doing tensile tests is that we observed that at high strains, the strain was not increasing linearly anymore (Figure 3.6 (A)). This is due to the fact that the non-linear part of the geometry started to deform (Figure 3.6 (B)).



(A) Non-linear strain at high strain

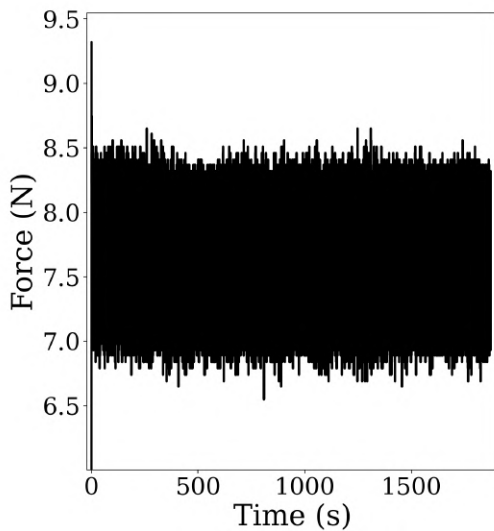


(B) Deformation of non-linear part of the geometry

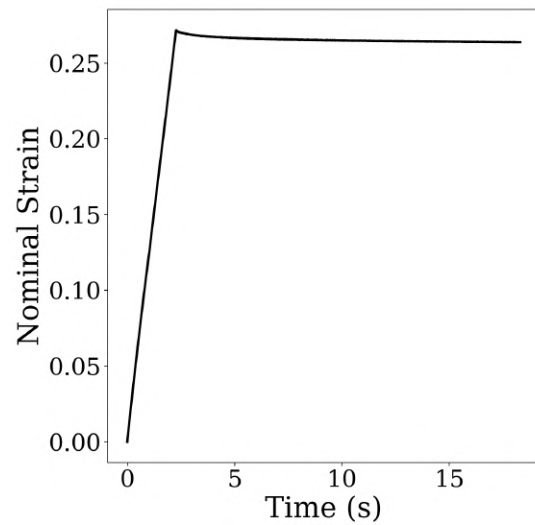
FIGURE 3.6: Experimental problems with tensile tests at high strain.

3.2.5 Relaxation Testing

For the relaxation testing, we used the geometry ② (Figure 3.2) and the experimental setup (A) (Figure 3.4). We used a load cell of 1500 N. We observed the following results:



(A) Noisy measurement



(B) Relaxation in strain

FIGURE 3.7: Experimental problems with relaxation.

We saw that we have a lot of noise on our force (± 0.75 N) which makes it impossible to fit and use this data. Furthermore, we see a decay in the strain measured. These are two problems we had to solve.

First, we increased the cross section of the geometry by multiplying the thickness by 4 (geometry ③ [Figure 3.2](#)). Then, we changed the load cell to a 200N load cell with an error of 0.1% (0.2 N). Since the load cell was smaller we had to redesign the setup, to be able to connect the grips to the load cell ([Figure 3.8](#)). So now, we are using the geometry ③ with the experimental setup (C) ([Figure 3.4](#)).

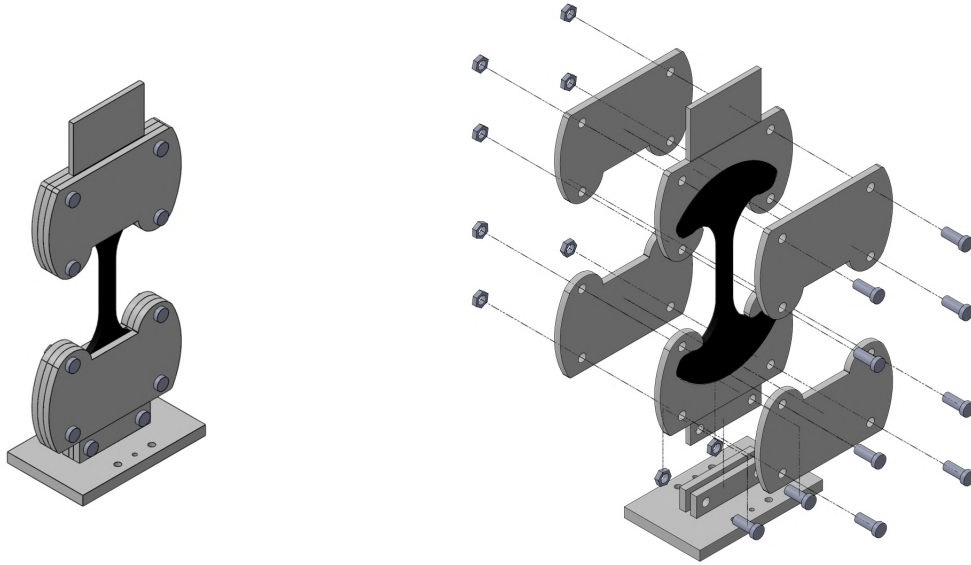


FIGURE 3.8: Exploded view of the relaxation setup with grips.

Nonetheless, we still got the decay in the nominal strain, which could be due to the use of grips. To solve this problem we tried doing relaxation tests with a dogbone geometry (ASTM D638 Type IV) and clamps (setup (D) [Figure 3.4](#)). In order to make sure that the sample has the same fixation force on both sides, we measured the distance between the clamps and chose 3 mm to be the most convenient. For this setup, we obtain a perfectly constant strain:

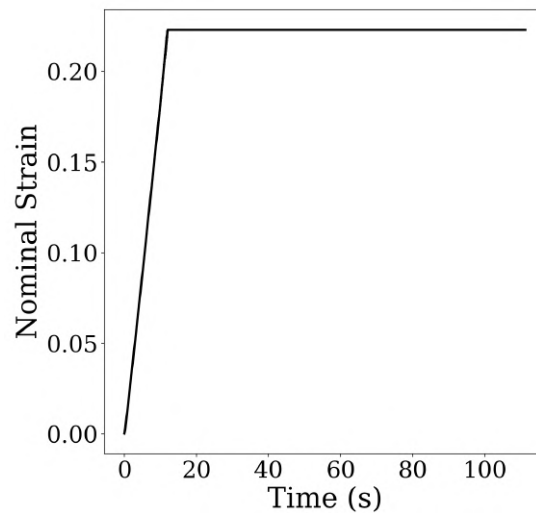


FIGURE 3.9: Nominal Strain as a function of time for relaxation of a dogbone sample.

3.3 Results and discussion

In this section, we will present the results we obtained from the previous mechanical testing: Tensile, Cyclic and Relaxation.

3.3.1 Tensile Test

After realizing tensile tests for different strain rates, we plot the engineering stress as a function of the engineering strain. We get the following plot:

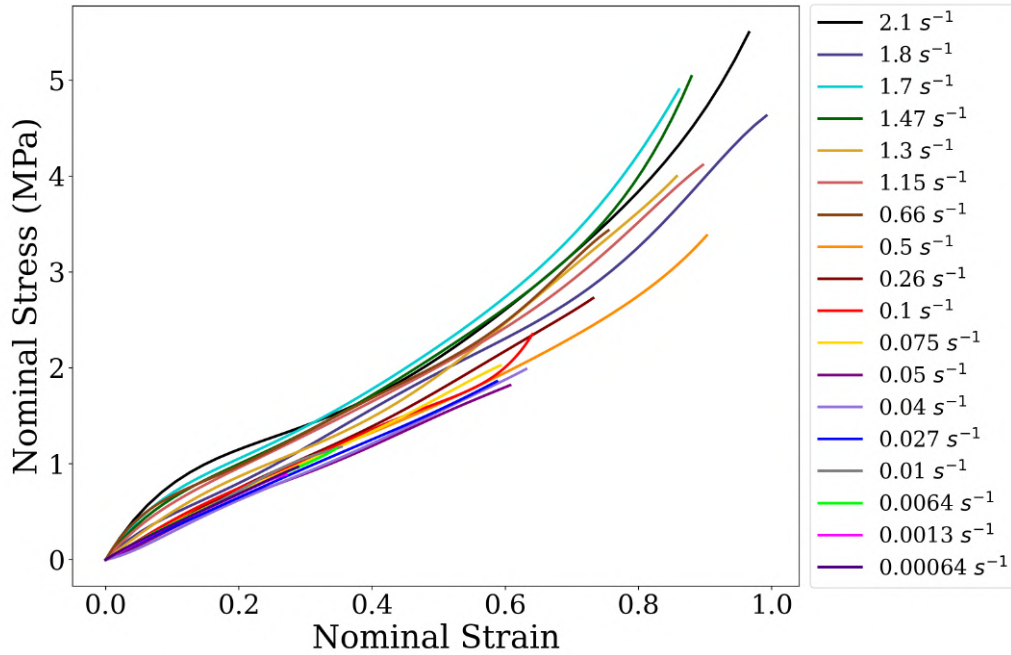


FIGURE 3.10: Stress-strain curves of the tensile tests of Tango Black for different strain rates.

These tests allow us to observe the rate-dependent behaviour of the material. In fact, we see that the stress response tends to be higher for higher strain rates. Furthermore, higher strain rates tend to lead to higher fracture strain (Section 3.5). This means that at higher strain rates, the material becomes more ductile and allows it to handle higher stresses.

Furthermore, now we get an idea of what is the maximal force that the sample can handle for different strain rates. We will use this information in our loading-unloading tests to set the maximal force for different strain rates.

Now, we fit the first linear part of the stress-strain curves to get the Young's modulus for different strain rates. We also find the Poisson Ratio by doing the ratio of the lateral and transverse strain. We get the following table:

TABLE 3.2: The Young's Modulus and Poisson Ratio of Tango Black for different strain rates.

Strain Rate	Young's Modulus	Poisson Ratio
10^{-1}	4.1	0.46
10^{-2}	4.007	0.4766
10^{-3}	3.914	0.4879
10^{-4}	3.818	0.4956

We observe that as the strain rate increases, the material becomes stiffer and the Poisson Ratio smaller.

3.3.2 Cyclic Test

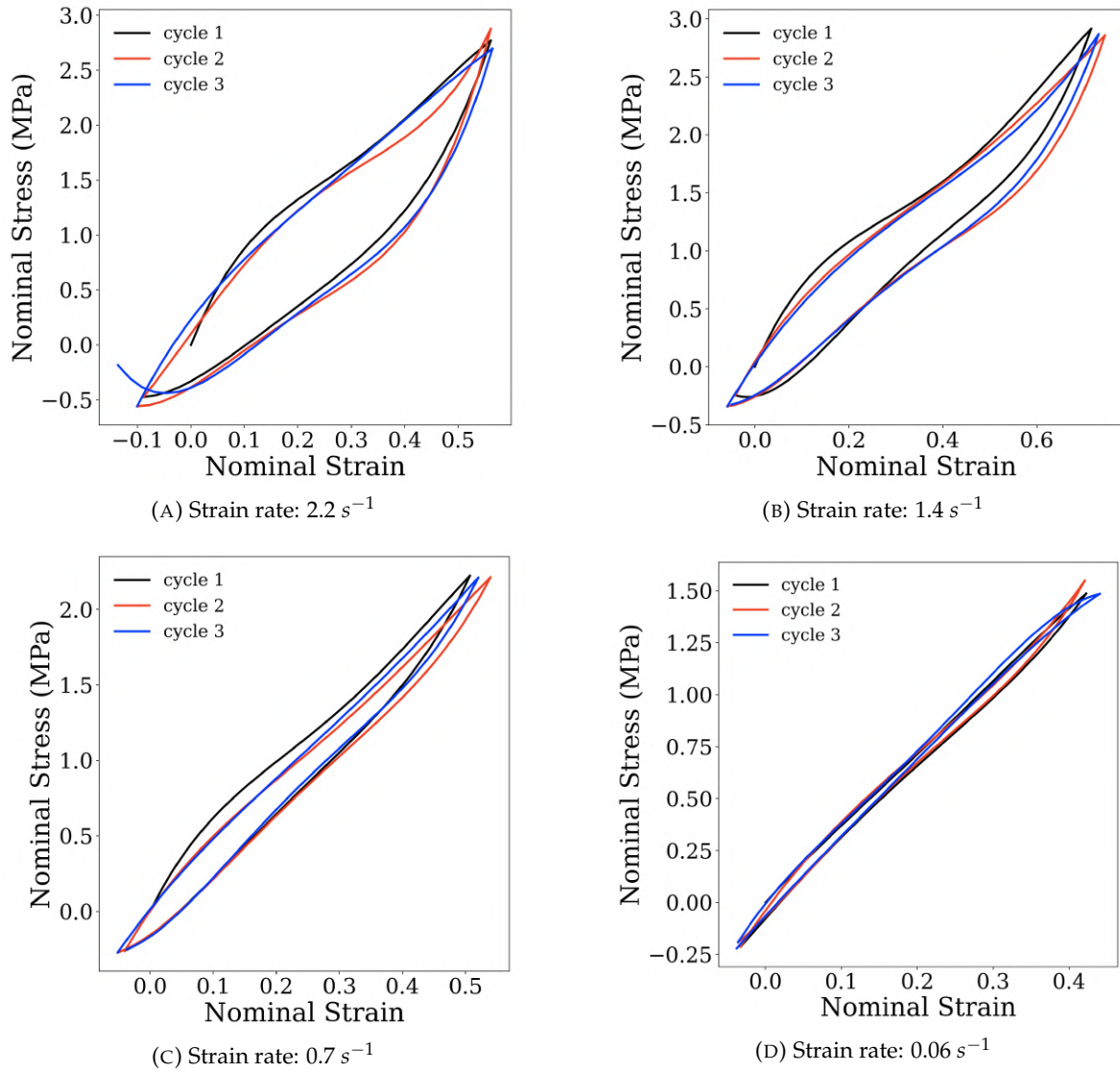


FIGURE 3.11: Three cycles of loading-unloading experiments at various strain rates.

The loading–unloading cyclic tests are typical in the case of mechanical characterization of the viscoelastic behaviour of a polymer. They demonstrate the rate-dependent behaviour of the material. In this case, the cyclic tests are carried out at different strain rates up to a maximal stress the sample can handle without breaking. We remark that for Tango Black strain rates between $0.1\text{--}2 \text{ s}^{-1}$ reveal best the dissipative behaviour, with larger hysteresis loops for larger strain rates. Tests are carried out at different strain rates: 0.0001 s^{-1} , 0.0064 s^{-1} , 0.01 s^{-1} , 0.06 s^{-1} , 0.1 s^{-1} , 0.7 s^{-1} , 1.4 s^{-1} and 2.2 s^{-1} .

The obtained results are presented in Figure 3.11 and 3.12 in terms of nominal stress versus nominal strain. Starting from the strain rate 0.06 s^{-1} , a rate-dependent non-linear behavior can be clearly observed in the loading and unloading of all the experiments (Figure 3.11). This could be due to the viscous nature of the material. Moreover, this data confirms that the viscous parameter of the material is non-linear and rate-dependent. We remark that the area inside of the hysteresis curves that represents the amount of dissipated energy per cycle gets larger as we increase the strain rate from 0.06 s^{-1} to 2.2 s^{-1} . This is the expected result for a viscoelastic material with rate-dependent behaviour. Finally, we can clearly see an increasing residual strain (amount of deformation which is not fully recovered after a full cycle) as we increase the strain rate

in Figure 3.12. These results are coherent with hyperelastic elastomer behaviour like VHB 1940 polymer [4].

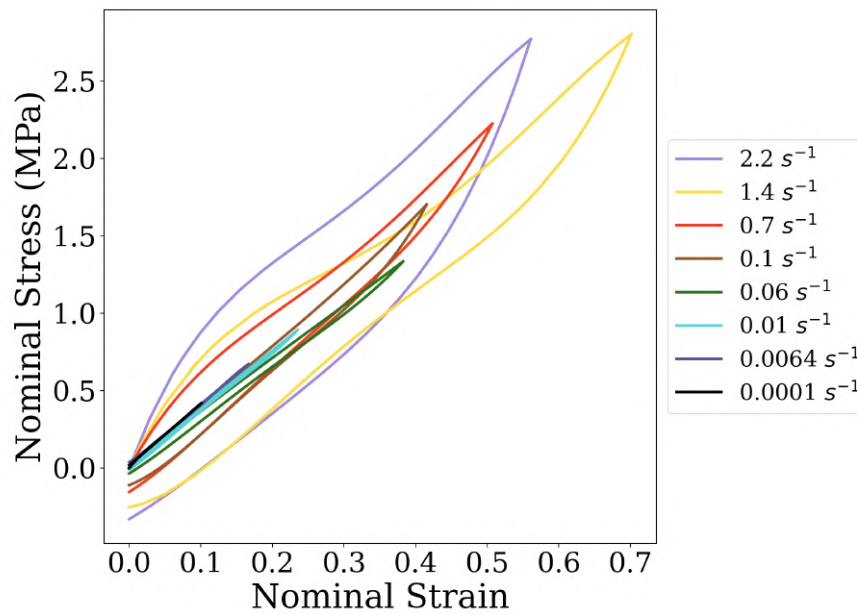


FIGURE 3.12: One cycle of loading-unloading experiments at various strain rates (0.0001 s^{-1} , 2.2 s^{-1})

3.3.3 Relaxation Test

Single-step relaxation at different stretch levels is another frequently used test to observe the equilibrium stress response, which is time-independent, as well as the rate-dependent viscous behaviour of a material. In fact, the higher the strain rate, the higher the initial value of the relaxation test. We realize the single-relaxation tests at different stretch levels. These tests were tricky to perform given the semi-brittle nature of Tango Black. In fact, stretching a sample at a relatively high strain (above 30%) would lead to a relaxation of a few seconds before observing the fracture of the sample (Section 3.5).

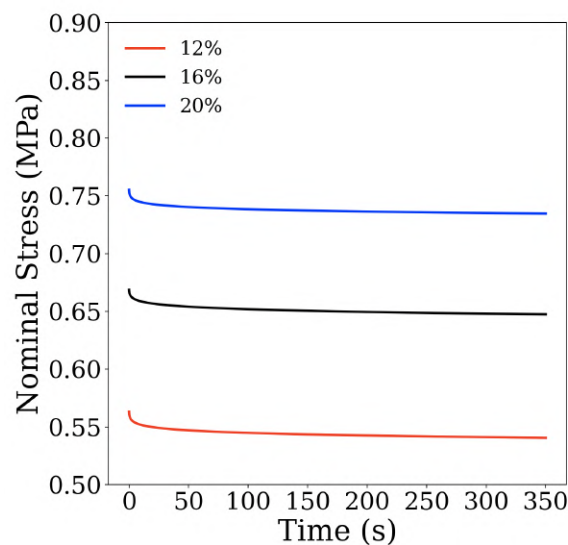


FIGURE 3.13: Single-step relaxation tests at 12%, 16%, 20% strains.

When applying the load on the sample, we used a displacement rate of 100 mm s^{-1} . We measured the stress response for 30 minutes for all the performed tests. We remark that after around 350 seconds the stress has reached a relatively constant value, so we will be presenting only the first 350 seconds of our tests in [Figure 3.13](#). We see that most of the decay occurs in the first 50 seconds of the test. We observe that as expected, the equilibrium stress response (i.e. the last value measured) is higher for higher strains. These results are coherent with hyperelastic elastomer behavior like VHB 1940 polymer [4].

3.4 Constitutive modeling and simulation

After realizing the tensile, cyclic and relaxation testing of Tango Black, we tried fitting the data with hyperelastic material models. We know that Tango Black is an incompressible elastomer since its Poisson Ratio is approximately 0.49. So we tried fitting first with a simple Neo-Hookean incompressible model with stress-deformation response:

$$S = \mu_1 \left(\lambda - \frac{1}{\lambda^2} \right) + \mu_2 \left(\frac{\lambda}{\lambda_v^2} - \frac{\lambda_v}{\lambda^2} \right) \quad (3.1)$$

And the following ODE in λ_v :

$$\dot{\lambda}_v = \frac{\mu_2}{3\eta_k} \left(\frac{\lambda^3 - \lambda_v^3}{\lambda \lambda_v} \right) \quad (3.2)$$

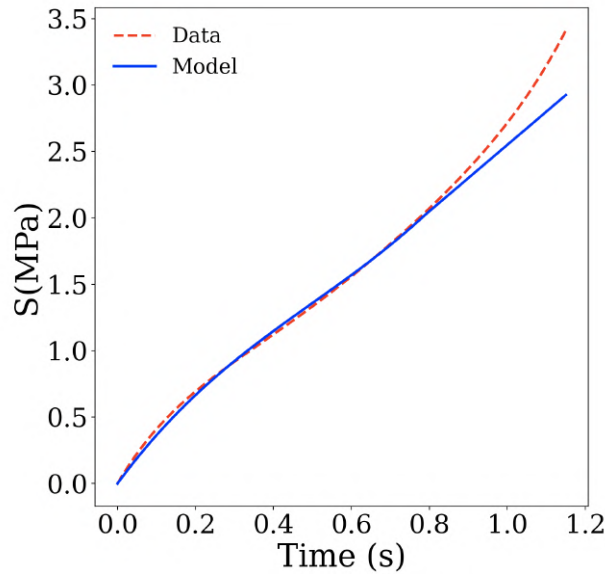


FIGURE 3.14: Neo-Hookean Incompressible Model Fitting of a tensile test at 0.66 s^{-1} .

We observe that this model fails to fit the hardening part of the curve given that we would need some power in the energy functions to fit such a sharp increase in stress. Now, we try to fit the data using the model for isotropic incompressible elastomers proposed by Lopez-Pamies [7]. We implement it using MATLAB. This model contains 14 parameters to fit our data. We start by solving the equation 2.12 using the Runge-Kutta Method ([Figure 4 Appendix C](#)). Then, we optimize the parameters in order to minimize the squared difference between the experimental data and the stress function (Equation 2.11)

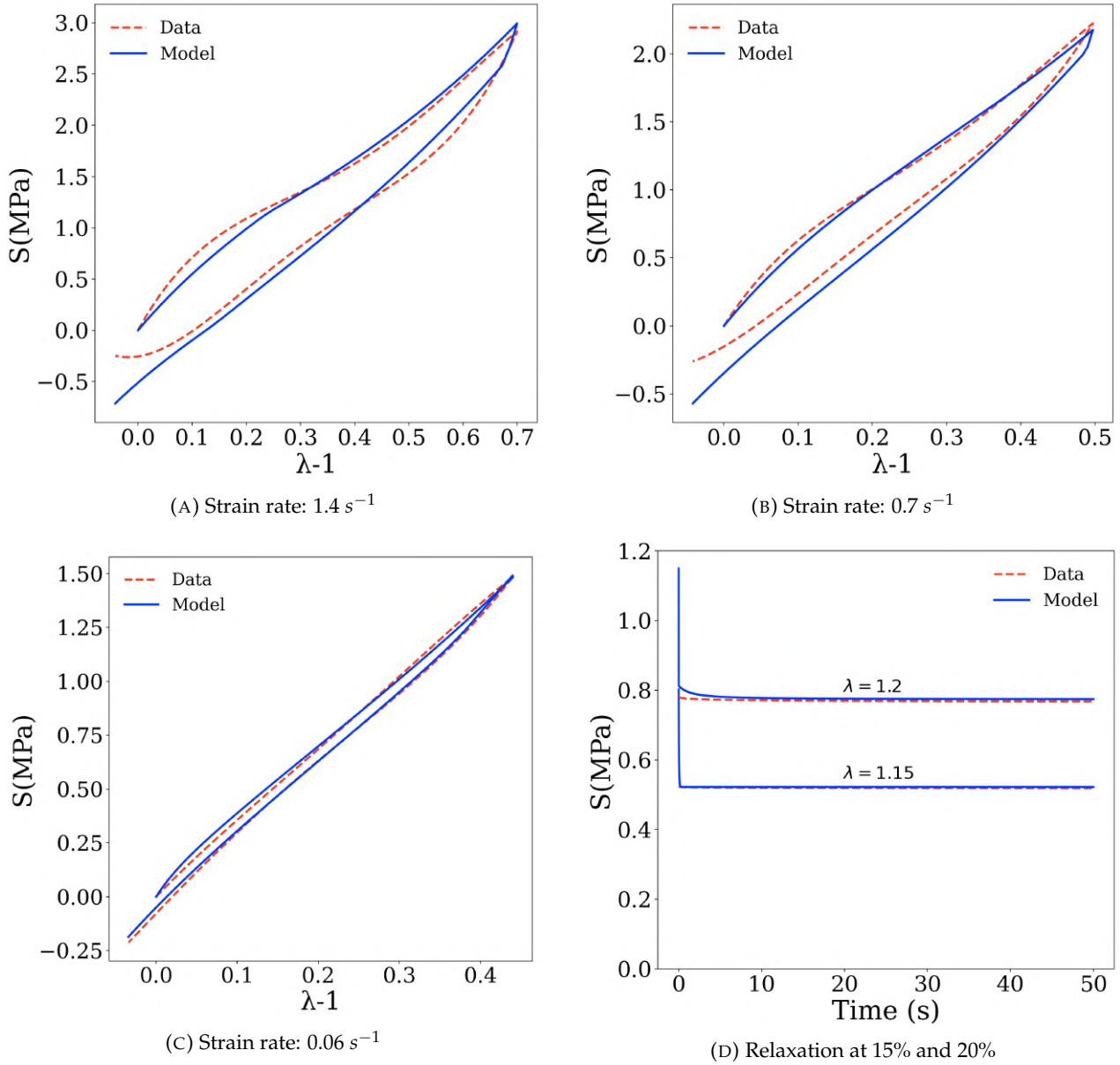


FIGURE 3.15: Fitting of the data from mechanical testing of Tango Black using the model proposed by Lopez-Pamies [7].

The optimization gives the following material parameters with a squared difference between the data and the model of 3.56.

TABLE 3.3: Material Parameters for Tango Black

$\mu_1 = 625.8 \text{ kPa}$	$\mu_2 = 625.8 \text{ kPa}$	$\alpha_1 = 2.3664$	$\alpha_2 = 2.3668$	
$m_1 = 362.8 \text{ kPa}$	$m_2 = 362.7 \text{ kPa}$	$a_1 = -1.3595$	$a_2 = -1.36$	$K_1 = 274 \text{ kPa s}$
$\eta_0 = 4.7 \text{ kPa s}$	$\eta_\infty = 0.3 \text{ kPa s}$	$\beta_1 = 26.7236$	$\beta_2 = 0$	$K_2 = 18.9752 \text{ MPa}^2$

Now that we have the parameters of Tango Black, we are interested in seeing how Tango Black behaves when we increase the strain rate applied at loading (Figure 3.16). We see that the stress increases with increasing strain rate. We have as well that the Young's modulus increases with increasing strain rate until reaching a maximal values that we can see at 100 s^{-1} and 1000 s^{-1} .

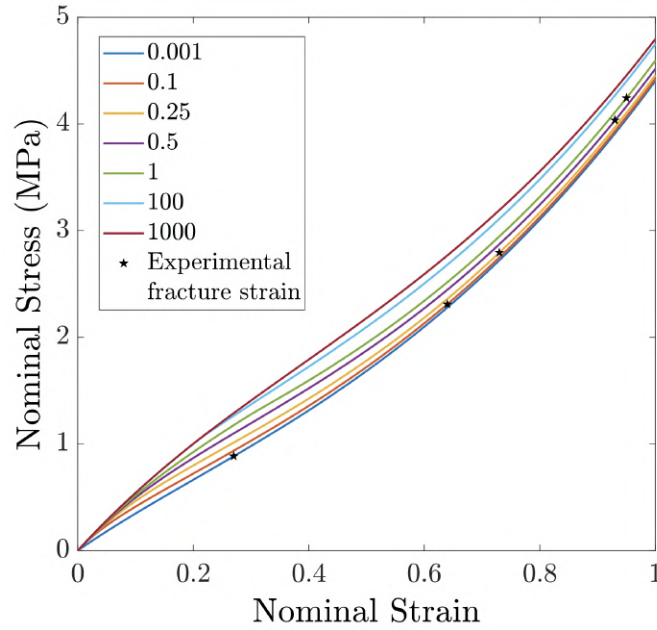


FIGURE 3.16: Simulation of tensile tests for different strain rates using the previous model.

3.5 Fracture Study

Given its semi-brittle nature, Tango Black manifests a very interesting failure behaviour. We see that its fracture stress and strain depends on the strain rate at which the tensile tests occurred. The higher the strain rate, the higher the strain and stress reached. This is due to the viscous nature of Tango Black, which allows it to deform plastically to reach higher strains.

Now, we plot the logarithm of the fracture stress as a function of the logarithm of the fracture strain in percent (Figure 3.17). We observe that we have a unique failure envelope as T. L. Smith states [8]. The deviation is at most 0.03 logarithmic unit.

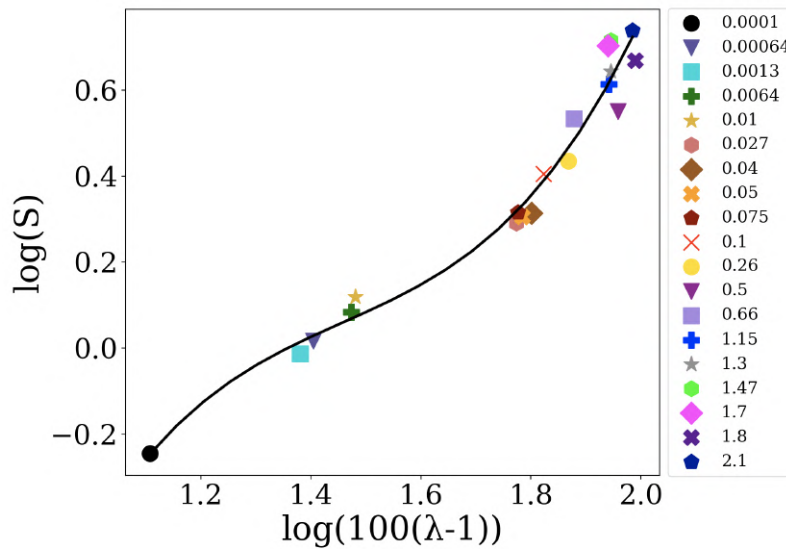


FIGURE 3.17: Fit of the log-log plot of the fracture stress as a function of the fracture strain.

Nonetheless, when we plot the fracture stress as a function of the fracture strain (Figure 3.18), we observe that as the stress and strain become bigger, the points deviate more from the envelope. The deviation

is at most 0.1 strain unit. So, it is the use of the log scale that lead to the conclusion of the existence of such envelope, but it remains a relatively good approximation of the fracture behavior.

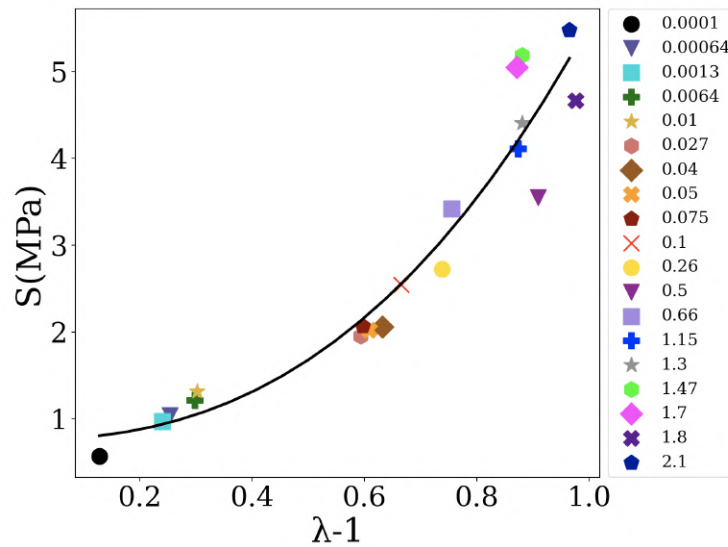


FIGURE 3.18: Fit of the plot of the fracture stress as a function of the fracture strain.

Finally, when we realized single-step relaxation tests at a relatively high strain (above 30%), we observe that our sample breaks after a few seconds. This is due to the existence of this fracture envelope. In fact, when we deform our sample at a given strain rate and stop at 30% strain, we have that the current strain rate becomes null, and thus, the sample is under a much higher stress than the one it can handle. Therefore, the sample ends up breaking, since the strain imposed is much higher than the one it would have reached under a quasi-static (small strain rate) tensile test ($\approx 16\%$).

4 Application: Compression of Tango Black Hexagonal Structures

In this section, we are going to present the experimental procedure and the results of the compression of Tango Black structures with hexagonal holes at different porosity densities. The goal behind these tests is to compare the behavior of the frame with hexagonal holes to the M-Veronoi generated structures.

4.1 Experimental Setup

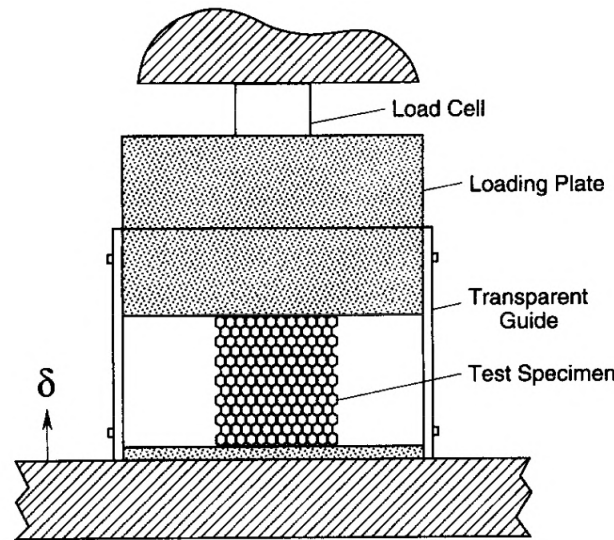


FIGURE 4.1: Schematic of test fixture used for in-plane compression of hexagonal structures [5].

The compression experiments were realized under displacement control in a hydraulic testing machine. The frames were placed between two transparent plates, preventing lateral buckling and were crushed by a loading plate made of Vero White (Figure 4.1). The guiding plates were lubricated to avoid contact friction from acting on the frames. For the data acquisition, we recorded the displacement applied to the sample from the initial position and the required load, using a load cell, every 0.1 seconds. The data was collected using the Software "Agnès". For these tests, we chose to compress the specimen at a displacement rate of 0.01mm/s. We stopped the experiments under force control. For a porosity of 70%, we stopped at 190N, for a porosity of 80%, we stopped at 120N, and for a porosity of 90%, we stopped at 10N.

A high resolution camera was used to capture pictures of the sample every second, in order to observe the collapse configurations of the frame at different points on the stress-strain curve.

4.2 Results and Conclusion

We plot the Engineering Stress as a function of the Engineering Strain and get the following graph:

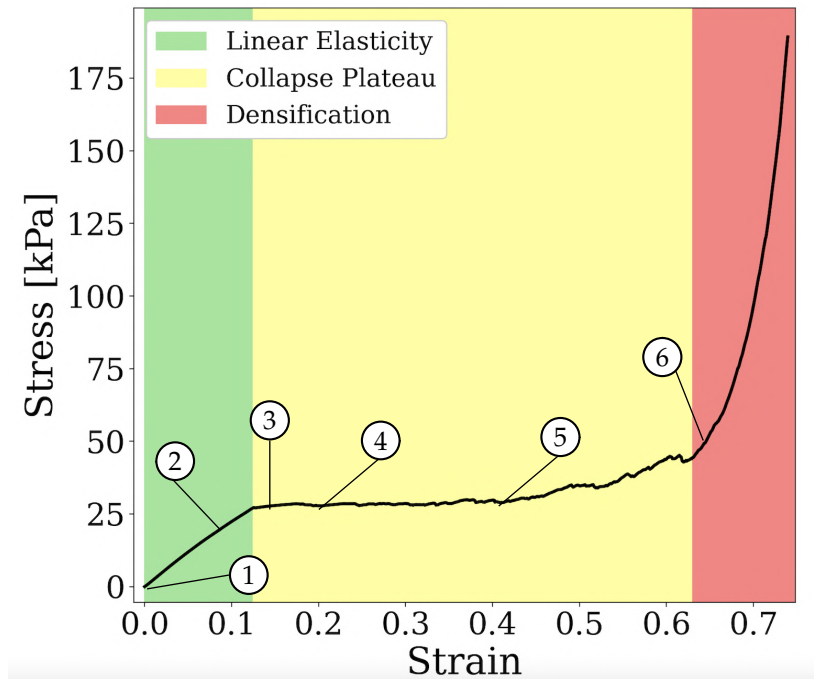


FIGURE 4.2: Schematic of compressive stress-strain curve for Tango Black hexagonal frame (porosity density 70%), showing the three regimes of linear elasticity, collapse plateau and densification.

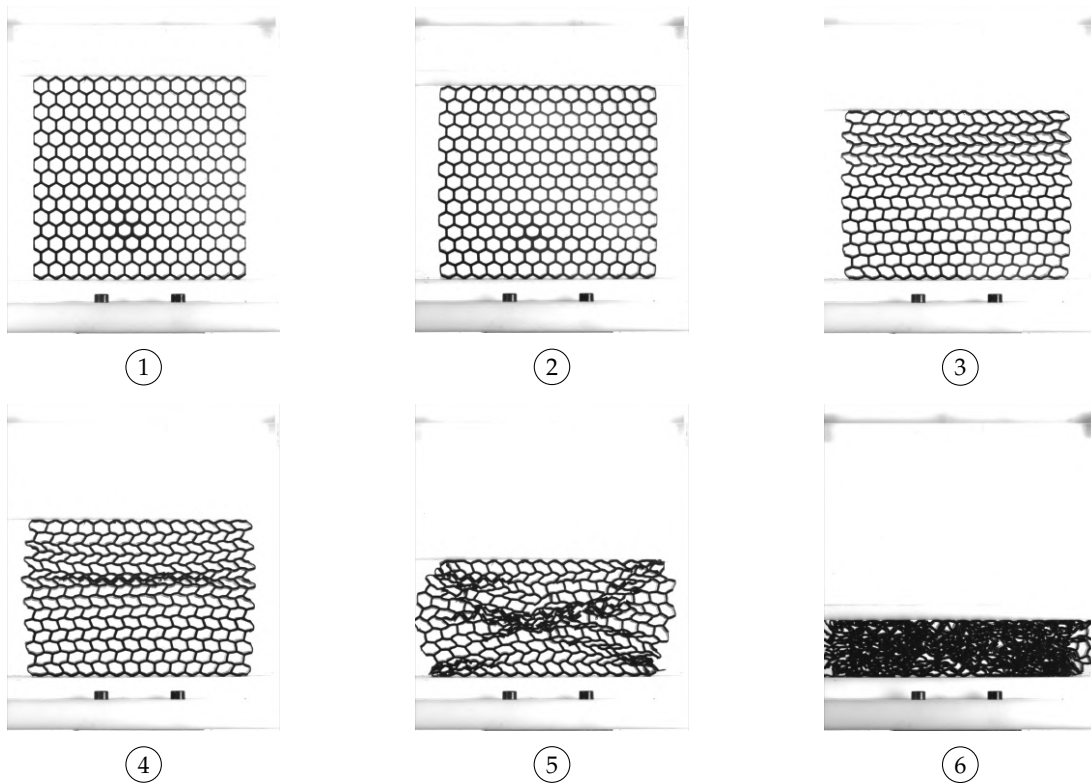


FIGURE 4.3: Sequence of collapse configurations of the hexagonal frame.

In [Figure 4.2](#), we see linear elasticity at low strains, followed by a long collapse plateau which lead to densification where the stress rises rapidly. This is the expected behavior knowing that these frames are comparable to elastic-brittle foams ([Figure 4.4](#)).

Linear elasticity, is characterized by the Young's modulus, where we observe symmetrical bending of the hexagons (see Figure 4.3 (2)). The collapse plateau is characterised by the buckling of hexagons as we can see in Figure 4.3 (3) which leads to a small drop of stress. Due to the brittle nature of Tango Black, we start to observe brittle fractures during the collapse plateau, (Figure 4.3 (4)) which keeps the stress constant for a long time. When the hexagons have almost been completely crushed, the material is compressed in a smaller and smaller volume (Figure 4.3 (6)), which gives this final area of rising stress.

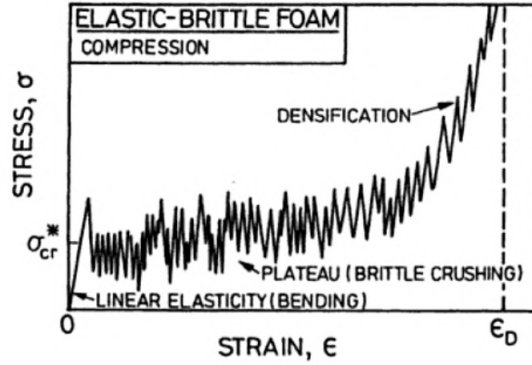


FIGURE 4.4: Schematic of compressive stress-strain curve for an elastic-brittle foam [6].

Now, we test Tango Black hexagonal frames with different porosities. We obtain the following curves:

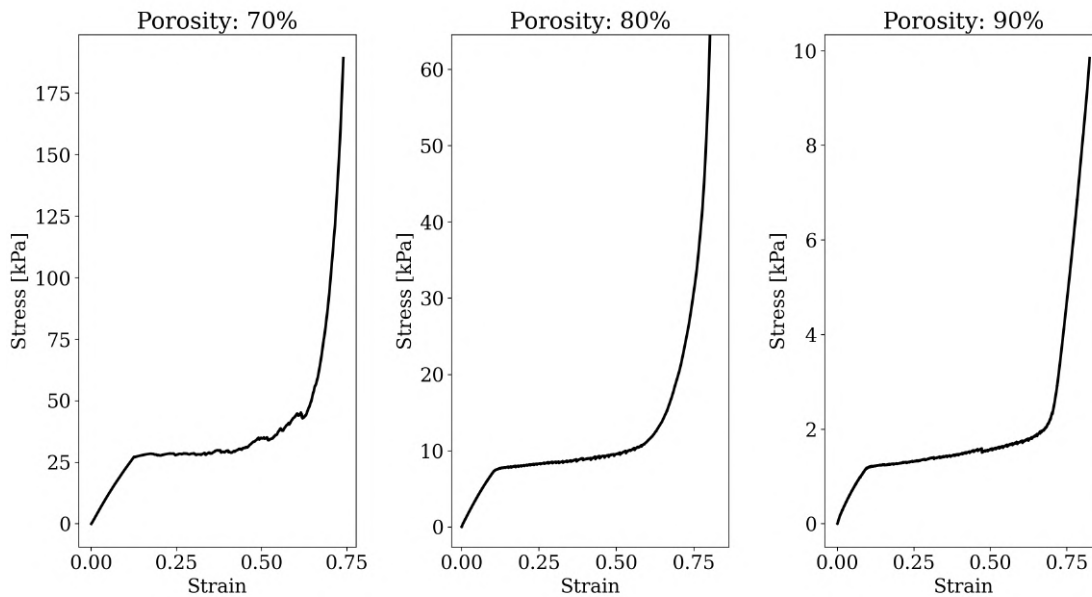


FIGURE 4.5: Stress-strain curves of Tango Black hexagonal frames with different porosity densities.

We observe that as we increase the porosity, the curve becomes smoother and smoother, which means that we start to observe less brittle fractures and more buckling in the collapse plateau. This is due to the loss of stiffness, which makes the material softer and easier to bend. We observe this loss of stiffness in the smaller slope of the linear part of the curve for higher porosity density.

Here are the values of the Young's modulus (E), the Plateau Stress (S_p) and the strain at which densification starts (ϵ_D) for different porosity densities (ρ):

TABLE 4.1: The Young's modulus, the Plateau Stress and the strain at which densification starts for different porosity densities.

ρ	E (kPa)	S_p (kPa)	ϵ_D
0.7	233.93 ± 5.66	30 ± 1.5	0.52 ± 0.05
0.8	97.21 ± 4.68	9 ± 1	0.6 ± 0.05
0.9	17.56 ± 2.37	1.6 ± 0.4	0.7 ± 0.05

We clearly see that decreasing the porosity density increases the Young's modulus and the Plateau Stress and reduces the strain at which densification starts. [6]

Finally, using the material parameters of Tango Black from [Section 3.4](#), it is possible to run a simulation that would fit the curves in [Figure 4.5](#) while considering the given hexagonal structure of the frames.

5 Conclusion

In conclusion, after realizing multiple mechanical experiments of tensile, loading-unloading and relaxation tests of a 3D printed polymer, Tango Black, I managed to fit the data with a model for isotropic incompressible elastomers proposed by Lopez-Pamies [7]. An application of the following work is to use this model with the parameters found in [Section 3.4](#) to run simulations of a given structure and compare it with the experimental data. An example of such application is given in [Section 4](#).

This research internship has been a real learning experience on all of the theoretical, experimental and numerical domains. First of all, I learned many notions in non-linear viscoelasticity, constitutive modeling and Solid Mechanics. It was an opportunity to look at Solid Mechanics from a deeper mathematical aspect since I took the course from an engineering perspective. I had also the chance of applying the knowledge I acquired during my Bachelor courses.

On the experimental aspect, it was a very challenging journey along which I faced many problems that I had to solve. It helped me develop my critical and analytical thinking. I had to redesign the experimental setup and the geometry depending on the tests that we were going to realize. I used SolidWorks to design all the parts I used. Furthermore, I got acquainted with many experimental methods like 3D printing, tensile tests and DIC (Digital Image Correlation).

Finally, on the numerical level, I worked on choosing the convenient model that fits the experimental data. I used the implementation of Zahra Hooshmand of the model proposed by Pamies-Lopez [7] on MATLAB to optimize the parameters of the model to fit the data. I, thus, strengthened my skills in MATLAB. Moreover, I developed my data analysis skills in Python to fit the experimental data.

Lastly, this internship at the Laboratoire de Mécanique des Solides has been a great research experience. I had to work on a big project and to tackle each step of the problem and go forward. I successfully completed the task of the constitutive modeling of Tango Black by mechanical testing. The next step would be to use this model in the simulations of the testing of structures and try to simulate its behaviour. It could be also interesting to compare the parameters of Tango Black with Tango Black Plus, another polymer of the family of Tango, which is more ductile than Tango Black.

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Appendices

A Appendix A: Derivation of the equations of the model for isotropic incompressible elastomers for the uni-axial loading case

We are interested in the uni-axial displacement control loading. The boundary conditions would be:

$$\mathbf{S} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & S_{33} \end{pmatrix} \quad \mathbf{F} = \begin{pmatrix} \frac{1}{\sqrt{\lambda}} & 0 & 0 \\ 0 & \frac{1}{\sqrt{\lambda}} & 0 \\ 0 & 0 & \lambda \end{pmatrix} \quad \mathbf{F}^v = \begin{pmatrix} \frac{1}{\sqrt{\lambda_v}} & 0 & 0 \\ 0 & \frac{1}{\sqrt{\lambda_v}} & 0 \\ 0 & 0 & \lambda_v \end{pmatrix} \quad (1)$$

We have that the stress-deformation response is given by [7]:

$$\mathbf{S} = \left[\sum_{r=1}^2 3^{1-\alpha_r} \mu_r I_1^{\alpha_r-1} \right] \mathbf{F} + \left[\sum_{r=1}^2 3^{1-a_r} m_r \left(\mathbf{C} \cdot \mathbf{C}^{v-1} \right)^{a_r-1} \right] \mathbf{F} \mathbf{C}^{v-1} - p \mathbf{F}^{-T} \quad (2)$$

with $I_1 = \text{tr} \mathbf{C}$, where $\mathbf{C} = \mathbf{F}^T \mathbf{F}$ is the right Cauchy–Green deformation tensor and $\mathbf{C}^v = \mathbf{F}^{vT} \mathbf{F}^v$. And the ensuing evolution equation can be written as [7]:

$$\dot{\mathbf{C}}^v = \frac{\sum_{r=1}^2 3^{1-a_r} m_r \left(\mathbf{C} \cdot \mathbf{C}^{v-1} \right)^{a_r-1}}{3\eta_k} \left(\mathbf{C} - \frac{1}{3} \left(\mathbf{C} \cdot \mathbf{C}^{v-1} \right) \mathbf{C}^v \right) \quad (3)$$

where

$$\eta_k = \eta_\infty + \frac{\eta_0 - \eta_\infty + k_1 \left[I_1^{v\beta_1} - 3^{\beta_1} \right]}{1 + \left(k_2 J_2^{NEq} \right)^{\beta_2}} \quad (4)$$

and

$$J_2^{NEq} = \left(\frac{I_1^{e2}}{3} - I_2^e \right)^2 \left(\sum_{r=1}^2 3^{1-a_r} m_r I_1^{e a_r-1} \right)^2 \quad (5)$$

Then, we get:

$$\mathbf{C} = \begin{pmatrix} \frac{1}{\lambda} & 0 & 0 \\ 0 & \frac{1}{\lambda} & 0 \\ 0 & 0 & \lambda^2 \end{pmatrix} \quad \mathbf{C}^v = \begin{pmatrix} \frac{1}{\lambda_v} & 0 & 0 \\ 0 & \frac{1}{\lambda_v} & 0 \\ 0 & 0 & \lambda_v^2 \end{pmatrix} \quad \mathbf{C}^e = \mathbf{C} \mathbf{C}^{v-1} = \begin{pmatrix} \frac{\lambda_v}{\lambda} & 0 & 0 \\ 0 & \frac{\lambda_v}{\lambda} & 0 \\ 0 & 0 & \frac{\lambda^2}{\lambda_v^2} \end{pmatrix} \quad (6)$$

$$\mathbf{B}^v = \mathbf{F} \mathbf{F}^{vT} = \begin{pmatrix} \frac{1}{\lambda_v} & 0 & 0 \\ 0 & \frac{1}{\lambda_v} & 0 \\ 0 & 0 & \lambda_v^2 \end{pmatrix} \quad \dot{\mathbf{C}}^v = \begin{pmatrix} \frac{-\dot{\lambda}_v}{\lambda_v^2} & 0 & 0 \\ 0 & \frac{-\dot{\lambda}_v}{\lambda_v^2} & 0 \\ 0 & 0 & 2\dot{\lambda}_v \lambda_v \end{pmatrix} \quad (7)$$

$$I_1^e = \text{tr} \mathbf{C}^e = \frac{2\lambda_v}{\lambda} + \frac{\lambda^2}{\lambda_v^2} \quad \frac{I_1^{e2}}{3} - I_2^e = \frac{\text{tr}(\mathbf{C}^e)^2}{3} - \frac{\text{tr}(\mathbf{C}^{e2})}{2} = \frac{1}{3} \left(\frac{\lambda_v}{\lambda} - \frac{\lambda^2}{\lambda_v^2} \right) \quad (8)$$

$$I_1^v = \text{tr} \mathbf{B}^v = \frac{2}{\lambda_v} + \lambda_v^2 \quad I_1 = \text{tr} \mathbf{C} = \frac{2}{\lambda} + \lambda^2 \quad (9)$$

Then, replacing in (2), (3), (4) and (5), we get:

$$\mathbf{S} \mathbf{F}^T = \left[\sum_{r=1}^2 3^{1-\alpha_r} \mu_r I_1^{\alpha_r-1} \right] \mathbf{F} \mathbf{F}^T + \left[\sum_{r=1}^2 3^{1-a_r} m_r \left(\mathbf{C} \cdot \mathbf{C}^{v-1} \right)^{a_r-1} \right] \mathbf{F} \mathbf{C}^{v-1} \mathbf{F}^T - \mathbf{p} \quad (10)$$

Then, let $A = \sum_{r=1}^2 3^{1-\alpha_r} \mu_r I_1^{\alpha_r-1}$ and $B = \sum_{r=1}^2 3^{1-a_r} m_r \left(\frac{2\lambda_v}{\lambda} + \frac{\lambda^2}{\lambda_v^2} \right)^{a_r-1}$,

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \lambda S_{33} \end{pmatrix} = A \begin{pmatrix} \frac{1}{\lambda} & 0 & 0 \\ 0 & \frac{1}{\lambda} & 0 \\ 0 & 0 & \lambda^2 \end{pmatrix} + B \begin{pmatrix} \frac{\lambda_v}{\lambda} & 0 & 0 \\ 0 & \frac{\lambda_v}{\lambda} & 0 \\ 0 & 0 & \frac{\lambda^2}{\lambda_v^2} \end{pmatrix} - \mathbf{p} \quad (11)$$

Then, equating the first element of the matrices:

$$\mathbf{p} = \frac{A}{\lambda} + B \frac{\lambda_v}{\lambda} \quad (12)$$

Then, replacing in (11), and equating the last element of the matrices:

$$S = \left[\sum_{r=1}^2 3^{1-\alpha_r} \mu_r \left(\frac{2}{\lambda} + \lambda^2 \right)^{\alpha_r-1} \right] \left(\lambda - \frac{1}{\lambda^2} \right) + \left[\sum_{r=1}^2 3^{1-a_r} m_r \left(\frac{2\lambda_v}{\lambda} + \frac{\lambda^2}{\lambda_v^2} \right)^{a_r-1} \right] \left(\frac{\lambda}{\lambda_v^2} - \frac{\lambda_v}{\lambda^2} \right) \quad (13)$$

Then,

$$\dot{\mathbf{C}}^v = \frac{\sum_{r=1}^2 3^{1-a_r} m_r \left(\frac{2\lambda_v}{\lambda} + \frac{\lambda^2}{\lambda_v^2} \right)^{a_r-1}}{\eta_k} \begin{pmatrix} \frac{1}{3\lambda} - \frac{\lambda^2}{3\lambda_v^3} & 0 & 0 \\ 0 & \frac{1}{3\lambda} - \frac{\lambda^2}{3\lambda_v^3} & 0 \\ 0 & 0 & \frac{2\lambda^2}{3} - \frac{2\lambda_v^3}{3\lambda} \end{pmatrix} \quad (14)$$

Then, equating the elements of the matrices of both sides, we get:

$$2\dot{\lambda}_v \lambda_v = \frac{\sum_{r=1}^2 3^{1-a_r} m_r \left(\frac{2\lambda_v}{\lambda} + \frac{\lambda^2}{\lambda_v^2} \right)^{a_r-1}}{\eta_k} \left(\frac{2\lambda^2}{3} - \frac{2\lambda_v^3}{3\lambda} \right) \quad (15)$$

Then,

$$\dot{\lambda}_v = \frac{\sum_{r=1}^2 3^{1-a_r} m_r \left(\frac{2\lambda_v}{\lambda} + \frac{\lambda^2}{\lambda_v^2} \right)^{a_r-1}}{3\eta_k} \left(\frac{\lambda^2}{\lambda_v} - \frac{\lambda_v^2}{\lambda} \right) \quad (16)$$

Finally, we get the following stress function S along the vertical direction:

$$S = \left[\sum_{r=1}^2 3^{1-\alpha_r} \mu_r \left(\frac{2}{\lambda} + \lambda^2 \right)^{\alpha_r-1} \right] \left(\lambda - \frac{1}{\lambda^2} \right) + \left[\sum_{r=1}^2 3^{1-a_r} m_r \left(\frac{2\lambda_v}{\lambda} + \frac{\lambda^2}{\lambda_v^2} \right)^{a_r-1} \right] \left(\frac{\lambda}{\lambda_v^2} - \frac{\lambda_v}{\lambda^2} \right) \quad (17)$$

And the following ODE in λ_v :

$$\dot{\lambda}_v = \frac{\sum_{r=1}^2 3^{1-a_r} m_r \left(\frac{2\lambda_v}{\lambda} + \frac{\lambda^2}{\lambda_v^2} \right)^{a_r-1}}{3\eta_k} \left(\frac{\lambda^2}{\lambda_v} - \frac{\lambda_v^2}{\lambda} \right) \quad (18)$$

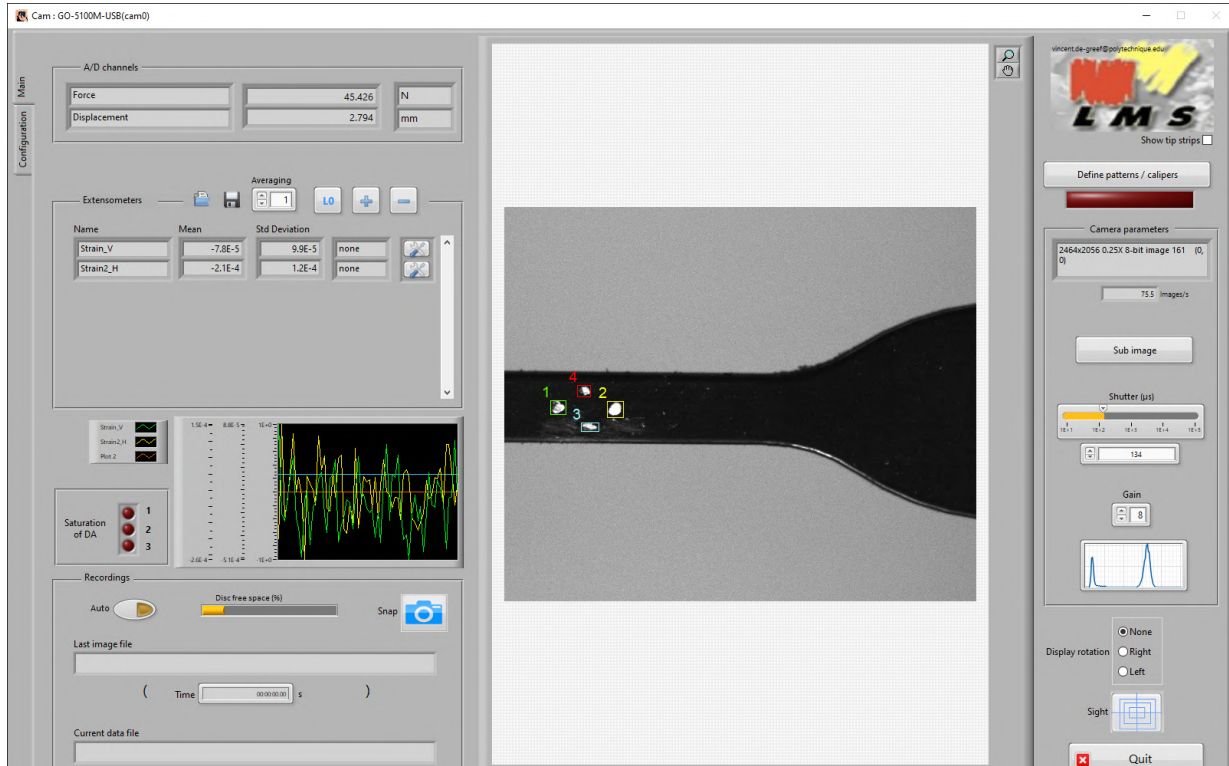
with

$$\eta_k = \eta_\infty + \frac{\eta_0 - \eta_\infty + k_1 \left[\left(\frac{2}{\lambda_v} + \lambda_v^2 \right)^{\beta_1} - 3^{\beta_1} \right]}{1 + \left(k_2 J_2^{NEq} \right)^{\beta_2}} \quad (19)$$

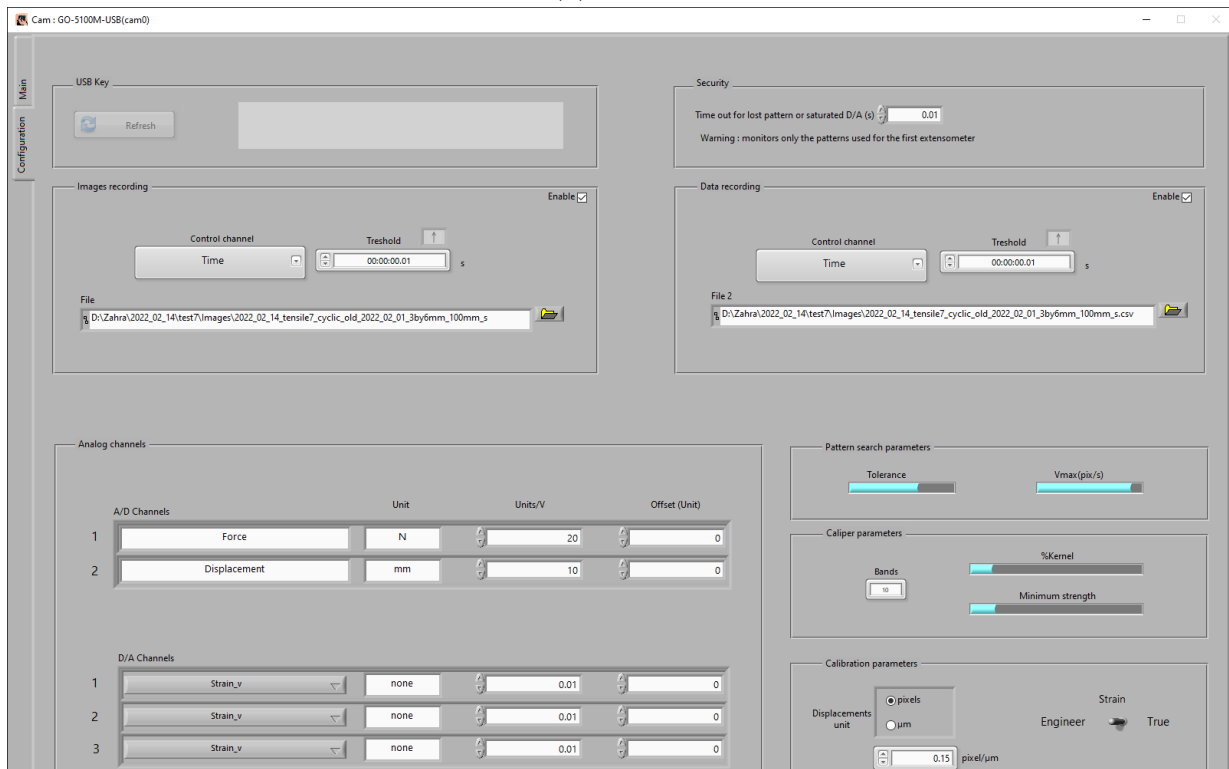
and

$$J_2^{NEq} = \frac{1}{3} \left(\frac{\lambda_v}{\lambda} - \frac{\lambda^2}{\lambda_v^2} \right)^2 \left(\sum_{r=1}^2 3^{1-a_r} m_r \left(\frac{2\lambda_v}{\lambda} + \frac{\lambda^2}{\lambda_v^2} \right)^{a_r-1} \right)^2 \quad (20)$$

B Appendix B: Softwares used

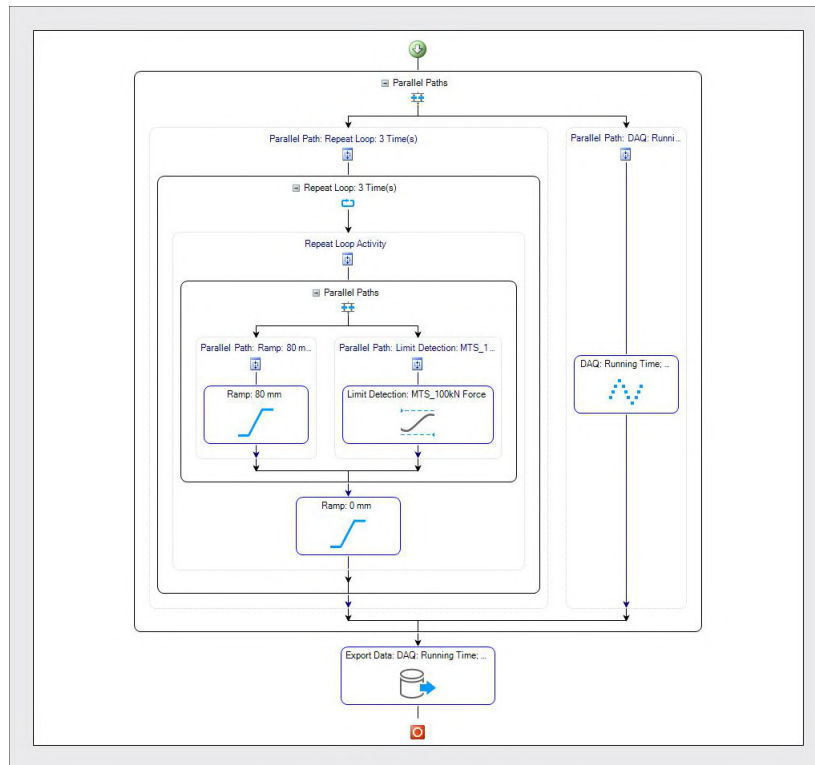


(A) Interface

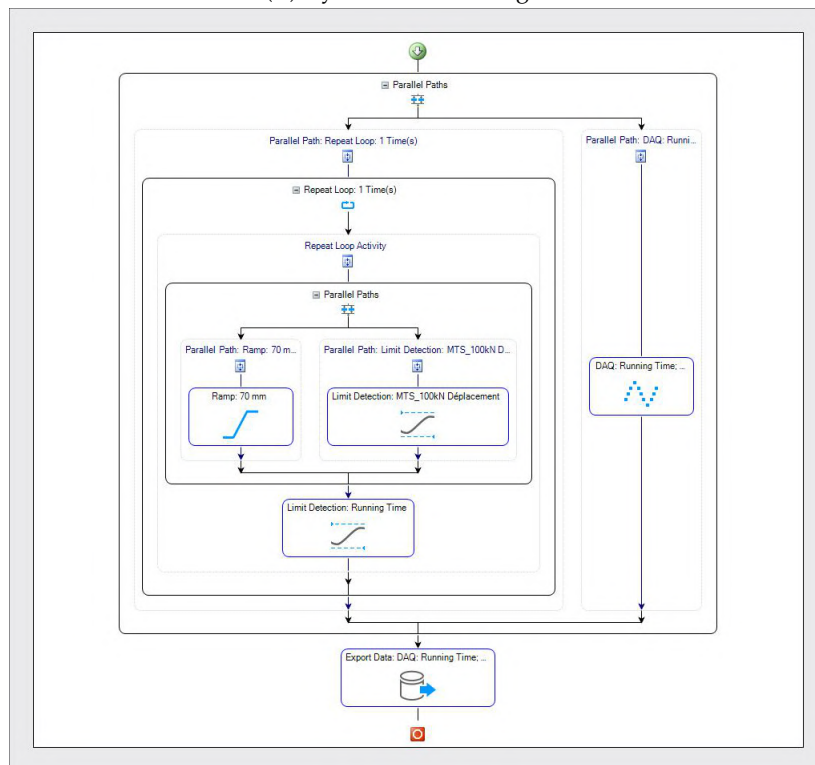


(B) Parameters

FIGURE 1: Sylvie IC: Camera Software.



(A) Cyclic MTS Test Program



(B) Relaxation MTS Test Program

FIGURE 2: Cyclic and Relaxation MTS TestSuite Multipurpose Elite Test Programs.

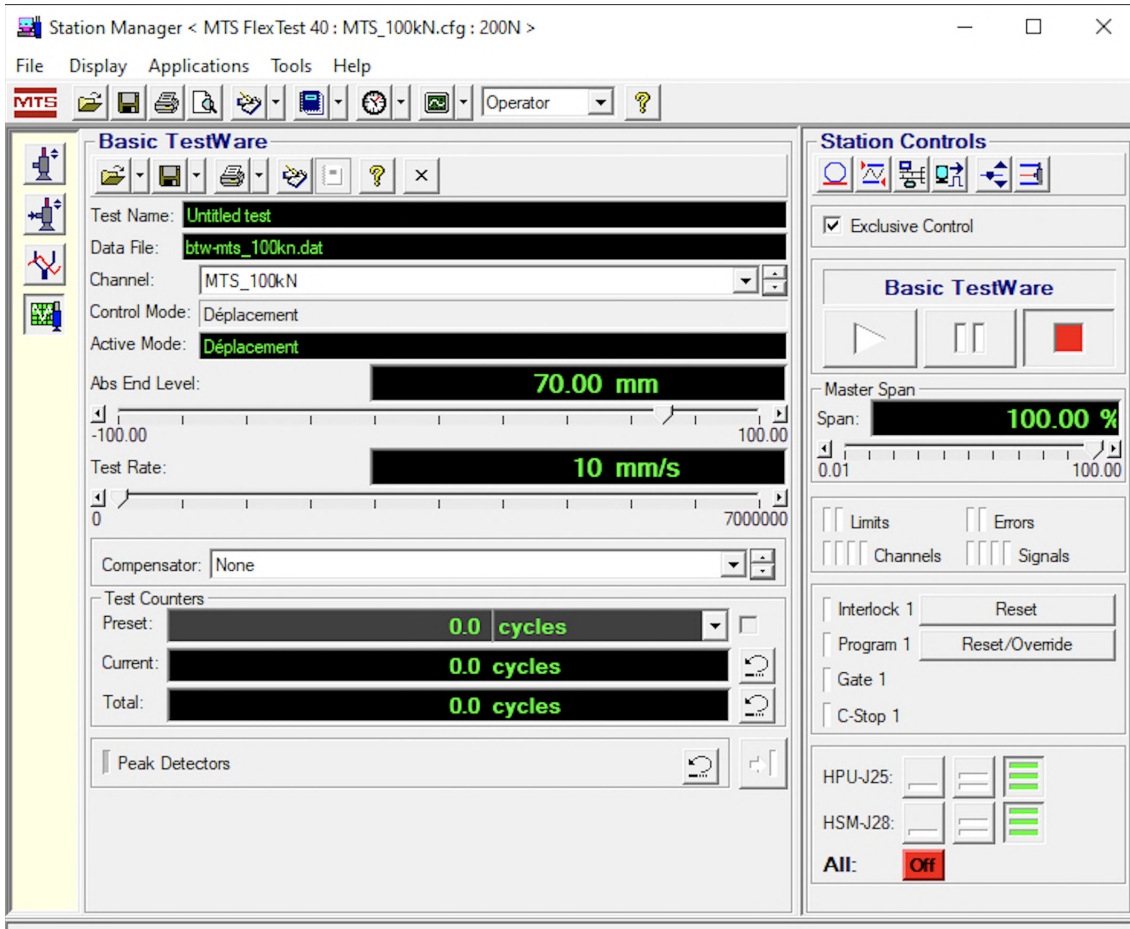


FIGURE 3: Station Manager Software.

C Appendix C: Runge-Kutta method implemented on MATLAB

```
function1[tt_] := (K1 = 0; K2 = 0; K3 = 0; K4 = 0; K5 = 0; K6 = 0;
    λ3v = 1; dttime = 1;
    tn = 0;
    While[tn < tt,
        K1 = g[tn, λ3v];
        K2 = g[tn + dttime/2, λ3v + dttime/2 K1];
        K3 = g[tn + dttime/4, λ3v + dttime/16 (3 K1 + K2)];
        K4 = g[tn + dttime/2, λ3v + dttime/16 (K3)];
        K5 = g[tn + 3 dttime/4, λ3v + dttime/16 3 (-K2 + 2 K3 + 3 K4)];
        K6 = g[tn + dttime, λ3v + dttime/7 (K1 + 4 K2 + 6 K3 - 12 K4 + 8 K5)];
        λ3v = λ3v + dttime/90 (7 K1 + 32 K3 + 12 K4 + 32 K5 + 7 K6); tn = tn + 1; S33num1 = S33[tt, λ3v]
```

FIGURE 4: Runge-Kutta method implemented on MATLAB.

D Appendix D: Images of samples used



(A) Geometry ①



(B) Geometry ②



(C) Geometry ③



(D) Dogbone

FIGURE 5: Images of samples used.



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